

Generative AI and LLMs

AI Learning Hub - Comprehensive Course Guide

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Generative Models Overview

Generative AI creates new content: text, images, audio, code.

Types of Generative Models:

- Autoencoders: Encode input to latent space, decode to reconstruct.
- VAEs (Variational Autoencoders): Learn probabilistic latent space.
- GANs (Generative Adversarial Networks): Generator vs Discriminator.
- Diffusion Models: Gradually add noise, learn to reverse the process.
Stable Diffusion, DALL-E use this approach for image generation.
- Autoregressive Models: Generate output token by token (GPT family).

Key applications: text generation, image synthesis, code completion, music generation, drug discovery, synthetic data generation.

Large Language Models

LLMs are transformer-based models trained on massive text corpora.

Training Pipeline:

1. Pre-training: Next token prediction on internet-scale data.
2. Supervised Fine-Tuning (SFT): Train on instruction-response pairs.
3. RLHF: Reinforcement Learning from Human Feedback.
 - Train reward model from human preferences.
 - Optimize policy using PPO against reward model.

Scaling Laws: Performance improves predictably with model size, data size, and compute. Emergent abilities appear at certain scales.

Inference Optimization: Quantization (INT8, INT4), KV-cache, speculative decoding, Flash Attention for efficient self-attention computation.

Prompt Engineering and RAG

Prompt Engineering techniques for better LLM outputs:

- Zero-shot: Direct instruction without examples.
- Few-shot: Provide examples in the prompt.
- Chain-of-Thought (CoT): 'Let's think step by step'.
- ReAct: Reasoning + Acting framework.

Retrieval-Augmented Generation (RAG):

1. Index documents: chunk text, create embeddings, store in vector DB.
2. Retrieve: Find relevant chunks using semantic similarity.
3. Generate: Feed retrieved context + query to LLM.

Benefits: Reduces hallucination, provides up-to-date knowledge, enables domain-specific responses without fine-tuning.

Vector Databases: FAISS, Pinecone, Weaviate, ChromaDB, Milvus.